On the most informative Boolean function under noise

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Abstract. Let X be a uniform random bit string and Y be obtained by flipping each bit of X independently with probability p. Let C be a random bit obtained as a function of Y. We show that

$$\mathbf{I}(C:X) \leq \mathbf{I}(Y_1:X)$$

for all $p \in [0,1]$, which resolves a conjecture posed by Courtade and Kumar in 2013.

1 Introduction

Let X be distributed uniformly on $\{0,1\}^n$ and let Y be obtained by flipping each coordinate of X with probability p, independently. In [12, 8], Courtade and Kumar ask: which bit C computed from Y is most informative about X? They conjecture that taking $C = Y_i$ for some $i \in [n]$ maximizes the information conveyed about X. In particular, their Conjecture 1 states that

$$\mathbf{I}(C:X) \leq \mathbf{I}(Y_1:X) = 1 - \mathrm{H}_2(p)\,. \tag{CK}$$

They further remark that, despite its resemblance to a simple homework exercise, the statement appears to involve substantial depth. Indeed, even after extensive efforts and partial progress [2, 3, 4, 7, 9, 10, 11, 14, 18, 19, 20, 22, 21, 23, 24, 25, 26], a full resolution of the conjecture remained elusive.

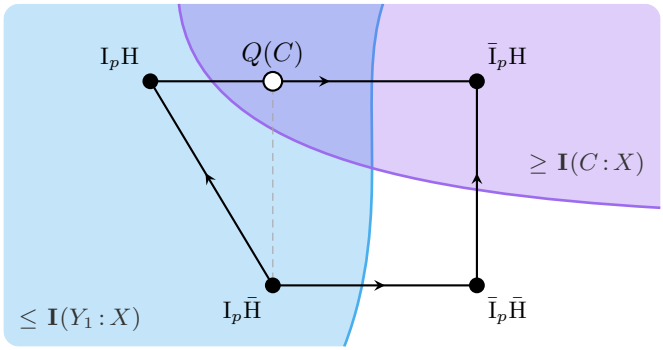


Figure 1. Arrows point toward the larger quantity. $Q(C)$ lies on the top edge, and its distance to $\bar{\mathrm{I}}_p\mathrm{H}(C)$ equals the length of the bottom edge, $(\bar{\mathrm{I}}_p - \mathrm{I}_p)\,\bar{\mathrm{H}}(C)$. Blue: quantities that are always at most $\mathbf{I}(Y_1:X)$. Purple: quantities that upper bound $\mathbf{I}(C:X)$. Only $Q(C)$ lies in both.

In this work, we observe that if we study the inequality parametrized by $\mu = \mathbb{E}[C]$, a delicate refinement of (CK) interpolating between Shannon and Tsallis entropies emerges. This refinement admits a proof by induction on the number of coordinates of X, Y. Denoting the Tsallis entropy of C by $\bar{\mathrm{H}}(C) = 4\mu(1-\mu)$ and setting $\mathrm{I}_p = \mathbf{I}(Y_1:X)$ and $\bar{\mathrm{I}}_p = (1-2p)^2$, we define the progressive entropy of C as

$$Q(C) = \bar{\mathrm{I}}_p\mathrm{H}(C) - (\bar{\mathrm{I}}_p - \mathrm{I}_p)\bar{\mathrm{H}}(C)\,. \tag{Q}$$

With this definition in hand, we prove the following.

Theorem 1.1. *Let X be a uniform random bit string and Y be obtained by flipping each bit of X with probability p, independently. For a bit C obtained from Y, we have*

$$\mathbf{I}(C:X) \leq Q(C) \leq \mathbf{I}(Y_1:X)$$

for any $p \in [0,1]$.

1.1 Comparison with earlier bounds

The strong data-processing inequality [1] gives $\mathbf{I}(C:X) \le \bar{\mathrm{I}}_p \mathrm{H}(C)$, the upper-right corner of Figure 1. Courtade and Kumar's [8] Conjecture 2 concerns the stronger upper bound $\mathrm{I}_p \mathrm{H}(C)$ at the upper-left corner. This bound was later shown to be too strong in general [5, 11]. Our refinement $Q(C)$ subtracts a Tsallis correction $(\bar{\mathrm{I}}_p - \mathrm{I}_p)\,\bar{\mathrm{H}}(C)$ from the strong data-processing bound and sits between the two.

1.2 Concurrent work

Concurrently with our work, Ky and Tran [13], Chen et al. [6], and Mahdavifar and Beirami [15] announced proofs of the Courtade–Kumar conjecture. Their preprints were first submitted to arXiv on September 21–22, 2026. We had obtained a proof using Fourier analysis by September 17, 2026, and completed writing down the simpler induction proof, which we present here, by September 20, 2026. Both versions are recorded in our public repository.[1]

OpenAI subsequently released two further proofs in its `openai/math` repository on October 6, 2026.[2] Both manuscripts are dated September 24, 2026. The first [16] derives (CK) from a Hellinger-contraction inequality and uses computer-assisted verification of scalar inequalities. The second [17] proves sharp information contraction for randomized binary outputs.

1.3 The continuous case

We note that $\mathbf{I}(C:X) \le Q(C)$ is a surprisingly sharp bound, which is already violated by a natural generalization of the problem. Let $X_\circ$ be distributed uniformly on the unit cube $[0,1]^n$ and $Y_\circ$ be obtained by independently resampling each coordinate of $X_\circ$ uniformly from $[0,1]$ with probability $2p$. Assume $C_\circ$ is a random bit obtained as a function of $Y_\circ$ with $\mu = \mathbb{E}\,[C_\circ]$.

Let T_μ be the indicator of $Y_{\circ,1} \le \mu$, so that $\mathbb{E}\,[T_\mu] = \mu$. In Lemma A.1 we show that $Q(C_\circ) < \mathbf{I}(T_\mu : X_\circ)$ for $0 < \mu < 1$, $\mu \ne 1/2$, and $0 < p < 1/2$, which shows that $Q(C_\circ)$ is too strong for the continuous problem. We conjecture that the threshold functions maximize mutual information at each fixed mean.

Conjecture 1.2. *Let $X_\circ$, $Y_\circ$ and $C_\circ$ be as constructed and T_μ be the indicator of the event $Y_{\circ,1} \le \mu$ as above. It holds that*

$$\mathbf{I}(C_\circ : X_\circ) \le \mathbf{I}(T_\mu : X_\circ)$$

for $0 \le p \le 1/2$ and $0 \le \mu \le 1$.

Note that a proof of Conjecture 1.2 directly implies (CK)—though not our Theorem 1.1—since given a strategy to output C from Y revealing $\mathbf{I}(C:X)$ bits of information, we can quantize $Y_\circ$ by rounding each coordinate to 0 or 1 with threshold $1/2$ and apply the strategy to obtain $C_\circ$. We get $\mathbf{I}(C_\circ : X_\circ) = \mathbf{I}(C:X)$ and it can be shown that $\mathbf{I}(T_\mu : X_\circ) \le \mathbf{I}(Y_1 : X)$.

2 Preliminaries

We use $\mathrm{H}(\cdot)$ and $\mathbf{I}(\cdot:\cdot)$ for Shannon entropy and mutual information, with logarithms in base 2. We denote the natural logarithm by ln. We only need Tsallis entropy for bits, while the observed variables may be arbitrary. For a bit C with mean μ, we define

$$\bar{\mathrm{H}}(C) = 4\mu(1-\mu).$$

For a bit C and an arbitrary random variable Z, Tsallis information is defined by

$$\bar{\mathbf{I}}(C\,;Z) = \bar{\mathrm{H}}(C) - \bar{\mathrm{H}}(C\,|\,Z)\,.$$

[1] The Fourier proof: `https://github.com/deepinfra/courtade-kumar/commit/1b7f244`. The induction proof: `https://github.com/deepinfra/courtade-kumar/commit/a07761c`.

[2] `https://openai.com/index/sharing-ai-progress-in-mathematics/`.

Note that for bits B, C, the quantities $\bar{\mathbf{I}}(B\,;C)$ and $\bar{\mathbf{I}}(C\,;B)$ need not be equal. We denote

$$\mathrm{I}_p = 1 - \mathrm{H}_2(p)\,, \qquad \bar{\mathrm{I}}_p = (1-2p)^2,$$
$$\text{and} \qquad \mathrm{J}_p = \bar{\mathrm{I}}_p - \mathrm{I}_p.$$

Thus $\mathrm{I}_p = \mathbf{I}(Y_i : X_i)$ and $\bar{\mathrm{I}}_p = \bar{\mathbf{I}}(Y_i\,; X_i)$. With these shorthands, we have

$$Q(C) = \bar{\mathrm{I}}_p \mathrm{H}(C) - \mathrm{J}_p \bar{\mathrm{H}}(C)\,.$$

For all our entropy and information quantities $\mathrm{H}(\cdot)$, $\mathbf{I}(\cdot:\cdot)$, $\bar{\mathrm{H}}(\cdot)$, $\bar{\mathbf{I}}(\cdot\,;\cdot)$, and Q, conditioning on U means evaluating the quantity under the conditional law given U and taking expectation over U, with p held fixed. For instance, $Q(C \,|\, U) = \mathbb{E}_{u\sim U}[Q(C \,|\, U = u)]$. Next, we gather some facts about Tsallis and Shannon information.

Lemma 2.1. *Let B and C be bits and Z, W, U be arbitrary random variables. We have*

1. $0 \le \bar{\mathbf{I}}(C\,; Z \,|\, U) \le \bar{\mathrm{H}}(C \,|\, U)$ *and* $\bar{\mathrm{H}}(C \,|\, U) \le \mathrm{H}(C \,|\, U) \le \big(1 + \bar{\mathrm{H}}(C \,|\, U)\big)/2$.
2. $\bar{\mathbf{I}}(C\,; Z, W \,|\, U) = \bar{\mathbf{I}}(C\,; Z \,|\, U) + \bar{\mathbf{I}}(C\,; W \,|\, Z, U)$. *If $C - Z - W$ is a Markov chain, then* $\bar{\mathbf{I}}(C\,; W) \le \bar{\mathbf{I}}(C\,; Z)$.
3. $\bar{\mathbf{I}}(C\,; Z \,|\, U) \le 2 \ln 2\, \mathbf{I}(C : Z \,|\, U)$. *If C is unbiased and independent of U, then also* $\mathbf{I}(C : Z \,|\, U) \le \bar{\mathbf{I}}(C\,; Z \,|\, U)$.
4. $\bar{\mathrm{H}}(C)\; \bar{\mathbf{I}}(B\,; C) = \bar{\mathrm{H}}(B)\; \bar{\mathbf{I}}(C\,; B)$. *If B is unbiased, then* $\bar{\mathbf{I}}(C\,; B) = \big(2 \Pr[B = C] - 1\big)^2$.
5. *If B is independent of U, then* $\bar{\mathbf{I}}(B\,; C \,|\, U)\; \bar{\mathrm{H}}(C \,|\, U) \ge \bar{\mathrm{H}}(B)\; \bar{\mathbf{I}}(C\,; B)$.
6. $0 \le \mathrm{J}_p \le \mathrm{I}_p$ *for every* $p \in [0, 1]$.

Proofs of the preceding and the next lemma are given in Appendix B. Let us define

$$c(\mu) = \frac{\ln(1/\mu)}{4(1-\mu)^2} \qquad (1/2 \le \mu < 1)\,,$$

extending it to $0 < \mu < 1/2$ by $c(\mu) = c(1-\mu)$.

Lemma 2.2. *For a nonconstant bit C with mean μ and every random variable U,*

$$\mathbf{I}(C : U) \le \frac{c(\mu)}{\ln 2}\, \bar{\mathbf{I}}(C\,; U)\,, \qquad \text{and} \qquad \bar{\mathrm{H}}(C)\; c(\mu) \le 1.$$

3 The proof

Our proof proceeds by induction on n, the number of coordinates of X, Y. Given the random bit C and the bit string Y, we adaptively pick a coordinate $i \in [n]$ and establish the bound

$$Q(C) - \mathbf{I}(C : X) \ge \frac{1}{1+\lambda}\, \big(Q(C \,|\, Y_i) - \mathbf{I}(C : X \,|\, Y_i)\big) \tag{$\star$}$$

for some $\lambda \ge 0$. When Y_i is fixed, C depends on $n-1$ coordinates and therefore the induction hypothesis applies for $Y_i = 0$ and $Y_i = 1$ separately, showing that $Q(C \,|\, Y_i) \ge \mathbf{I}(C : X \,|\, Y_i)$. Plugging into ($\star$), we get $Q(C) \ge \mathbf{I}(C : X)$, which implies (CK) as we show next.

Lemma 3.1. *For all $p \in [0, 1]$, we have $Q(C) \le \mathbf{I}(Y_1 : X)$.*

Proof. From Lemma 2.1(6) and (1), we have $\mathrm{J}_p \le \mathrm{I}_p$ and $\mathrm{H}(C) - \bar{\mathrm{H}}(C) \le 1 - \mathrm{H}(C)$. Let us write

$$\begin{aligned} Q(C) - \mathrm{I}_p \mathrm{H}(C) &= \mathrm{J}_p \mathrm{H}(C) - \mathrm{J}_p \bar{\mathrm{H}}(C) \\ &= \mathrm{J}_p \big(\mathrm{H}(C) - \bar{\mathrm{H}}(C)\big) \\ &\le \mathrm{I}_p \,(1 - \mathrm{H}(C))\,. \end{aligned}$$

Therefore $Q(C) \le \mathrm{I}_p\,(1 - \mathrm{H}(C)) + \mathrm{I}_p\, \mathrm{H}(C) = \mathbf{I}(Y_1 : X)$ as desired. □

The property we want from the coordinate i is fairly mild: $\bar{\mathbf{I}}(C\,;Y_i) \le 1/4$. The next lemma shows that such a coordinate always exists.

Lemma 3.2. *When $n \ge 2$, there exists an $i \in [n]$ such that $\bar{\mathbf{I}}(C\,;Y_i) \le 1/4$.*

Proof. Let $i, j \in [n]$ be distinct. We may relabel Y_i, Y_j so that $\mathbb{E}[C(2Y_i-1)]$ and $\mathbb{E}[C(2Y_j-1)]$ are nonnegative. Since Y_i, Y_j are independent uniform bits,

$$\sqrt{\bar{\mathbf{I}}(C\,;Y_i)} + \sqrt{\bar{\mathbf{I}}(C\,;Y_j)} = 2\,\mathbb{E}\left[C(2Y_i-1) + C(2Y_j-1)\right] \le 4\Pr[Y_i = Y_j = 1] = 1,$$

using $0 \le C \le 1$. At least one of the two square roots is at most $1/2$. □

Let us fix such an i satisfying $\bar{\mathbf{I}}(C\,;Y_i) \le 1/4$ and use the shorthands $A = X_i$, $B = Y_i$, and $U = X_{-i}$ in the rest of the paper. The following lemma is the crux of our argument, which establishes (⋆) giving the induction step.

Lemma 3.3. *Assuming $\bar{\mathbf{I}}(C\,;B) \le 1/4$, we have*

$$Q(C) - \mathbf{I}(C:X) \ge \frac{1}{1+\lambda}\left(Q(C\,|\,B) - \mathbf{I}(C:X\,|\,B)\right)$$

for some $\lambda \ge 0$.

Section 3.2 is devoted to proving Lemma 3.3; here we show that it implies the main theorem.

Theorem 1.1 (restated)**.** *It holds that $\mathbf{I}(C:X) \le Q(C) \le \mathbf{I}(Y_1:X)$.*

Proof. The second inequality is Lemma 3.1. For the first, the cases $p \in \{0, 1/2, 1\}$ and constant C are immediate. Complementing Y if necessary lets us assume $0 < p < 1/2$. We proceed by induction on n. When $n = 1$, the bit C is a constant, Y_1, or its negation, and the bound holds with equality. For $n \ge 2$, by Lemma 3.2 there exists a coordinate i with $\bar{\mathbf{I}}(C\,;Y_i) \le 1/4$. Conditioning on $B = Y_i$ reduces the problem to $n-1$ coordinates, so the induction hypothesis applies to the $B = 0$ and $B = 1$ cases individually, giving

$$\mathbf{I}(C:X\,|\,B) = \mathbf{I}(C:U\,|\,B) \le Q(C\,|\,B)\,.$$

Applying Lemma 3.3, we conclude $Q(C) - \mathbf{I}(C:X) \ge 0$. □

3.1 A comparison lemma

Let U be a random variable and A, B be unbiased bits jointly independent of U. Let C be a bit obtained from (B, U) using private randomness, with $\bar{\mathrm{H}}(C\,|\,U) > 0$. We compare C with an unbiased bit C' obtained from B alone using private randomness, subject to the matching condition

$$\mathbf{I}(B:C') = \frac{\mathbf{I}(B:C\,|\,U)}{\bar{\mathrm{H}}(C\,|\,U)}. \tag{B}$$

Lemma 3.4. *A bit C' satisfying (B) exists. Every such bit satisfies*

$$\mathbf{I}(A:C') \ge \frac{\mathbf{I}(A:C\,|\,U)}{\bar{\mathrm{H}}(C\,|\,U)}.$$

Proof. For the fixed channel between A and B, let F be the function determined by

$$F\left(\mathbf{I}(B:D)\right) = \mathbf{I}(A:D)$$

as D ranges over unbiased bits generated from B. It can be shown that F is increasing and concave (cf. Lemma A.2), and

$$\begin{aligned}
&\mathbf{I}(B:C\,|\,U=u) \le \bar{\mathrm{H}}(C\,|\,U=u)\,,\\
&\mathbf{I}(A:C\,|\,U=u) \le \bar{\mathrm{H}}(C\,|\,U=u)\,F\left(\frac{\mathbf{I}(B:C\,|\,U=u)}{\bar{\mathrm{H}}(C\,|\,U=u)}\right).
\end{aligned}$$

Values of u with zero conditional Tsallis entropy contribute zero to both informations and may be omitted. The first inequality, averaged over U, places the right-hand side of (B) in $[0,1]$. Varying the noise in the channel from B to C' therefore gives a matching bit. Applying Jensen's inequality to the second inequality, with weights proportional to $\bar{\mathrm{H}}(C\,|\,U=u)$, gives

$$\mathbf{I}(A:C\,|\,U) \le \bar{\mathrm{H}}(C\,|\,U)\, F\left(\frac{\mathbf{I}(B:C\,|\,U)}{\bar{\mathrm{H}}(C\,|\,U)}\right) = \bar{\mathrm{H}}(C\,|\,U)\,\mathbf{I}(A:C')\,,$$

as required. □

Lemma 3.5. *Every bit C' satisfying (B) also satisfies*

$$\bar{\mathbf{I}}(C';B) \ge \frac{\bar{\mathbf{I}}(C;B)}{\bar{\mathrm{H}}(C\,|\,U)^2}.$$

Proof. Let us write $v = \bar{\mathrm{H}}(C\,|\,U)$ and $d = \mathbb{E}[(2B-1)(2C-1)]$. Since B is unbiased, $\bar{\mathbf{I}}(C;B) = d^2$. As shown in Appendix A, $|d| \le v$ and

$$\frac{\mathbf{I}(B:C\,|\,U)}{v} \ge \mathbf{D}_2((1+d/v)/2\,\|\,1/2)\,.$$

If $\theta = \mathbb{E}[(2B-1)(2C'-1)]$, then $\bar{\mathbf{I}}(C';B) = \theta^2$ and $\mathbf{I}(B:C') = \mathbf{D}_2((1+\theta)/2\,\|\,1/2)$. The matching identity and monotonicity of this divergence in $|\theta|$ imply $\theta^2 \ge d^2/v^2$. □

3.2 The induction step

Recalling that $A = X_i$, $B = Y_i$ and $U = X_{-i}$, in this section we would like to lower bound the gap $Q(C) - \mathbf{I}(C:X)$ by a factor of the conditional gap

$$\Delta = Q(C\,|\,B) - \mathbf{I}(C:X\,|\,B)\,.$$

We obtain two lower bounds on $Q(C) - \mathbf{I}(C:X)$. The first involves Δ and benefits from C depending on U, through a positive term in $\bar{\mathbf{I}}(C;U)$. The second has a negative term in $\bar{\mathbf{I}}(C;U)$ and benefits from weaker dependence. We add λ times the second bound to the first, choosing λ to cancel these terms, so that the bound works whether or not C depends on U.

To analyze the remaining terms, we introduce an unbiased comparison bit C' obtained from B using private randomness. The comparisons in Lemma 3.4 and Lemma 3.5, together with our choice of coordinate with $\bar{\mathbf{I}}(C;B) \le 1/4$, allow us to charge these terms to quantities involving $\mathbf{I}(A:C')$ and $\mathbf{I}(B:C')$. What remains is an inequality about three unbiased bits forming the Markov chain $A - B - C'$, which we analyze in Section 3.3.

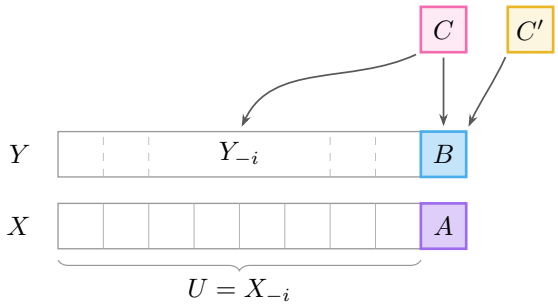


Figure 2. Arrows indicate dependence. The original bit C depends on Y_{-i} and B, while the comparison bit C' depends only on B and private randomness.

Let us start with a lemma that we use in obtaining the first lower bound.

Lemma 3.6. *If $\bar{\mathrm{H}}(C\,|\,U) > 0$, then*

$$\mathbf{I}(B:C\,|\,U) - \mathbf{I}(B:C) \ge \frac{1}{2\ln 2}\,\frac{\bar{\mathbf{I}}(C;B)\;\bar{\mathbf{I}}(C;U)}{\bar{\mathrm{H}}(C)\;\bar{\mathrm{H}}(C\,|\,U)}.$$

Proof. Since B and U are independent, the Shannon and Tsallis chain rules give

$$\begin{aligned}\mathbf{I}(B:C\,|\,U) - \mathbf{I}(B:C) &= \mathbf{I}(B:U\,|\,C)\,,\\ \bar{\mathbf{I}}(B\,;C\,|\,U) - \bar{\mathbf{I}}(B\,;C) &= \bar{\mathbf{I}}(B\,;U\,|\,C)\,.\end{aligned}$$

Since B is unbiased and independent of U, parts 3–5 of Lemma 2.1 give

$$\begin{aligned}\mathbf{I}(B:U\,|\,C) &\geq \frac{\bar{\mathbf{I}}(B\,;U\,|\,C)}{2\ln 2}\\ &\geq \frac{\bar{\mathbf{I}}(C\,;B)}{2\ln 2}\left[\frac{1}{\bar{\mathrm{H}}(C\,|\,U)} - \frac{1}{\bar{\mathrm{H}}(C)}\right]\\ &= \frac{1}{2\ln 2}\frac{\bar{\mathbf{I}}(C\,;B)\;\bar{\mathbf{I}}(C\,;U)}{\bar{\mathrm{H}}(C)\;\bar{\mathrm{H}}(C\,|\,U)},\end{aligned}$$

as required. □

When $\bar{\mathrm{H}}(C\,|\,U) > 0$, let C' be an unbiased bit obtained from B alone using private randomness and satisfying (B).

Lemma 3.7. *If* $\bar{\mathrm{H}}(C\,|\,U) > 0$*, then*

$$\begin{aligned}Q(C) - \mathbf{I}(C:X) \geq \Delta &+ \bar{\mathrm{H}}(C\,|\,U)\left(\bar{\mathrm{I}}_p\mathbf{I}(B:C') - \mathbf{I}(A:C')\right)\\ &+ \frac{(1-\bar{\mathrm{I}}_p)\bar{\mathbf{I}}(C\,;B)\;\bar{\mathbf{I}}(C\,;U)}{2\ln 2\,\bar{\mathrm{H}}(C)\;\bar{\mathrm{H}}(C\,|\,U)} - \mathrm{J}_p\bar{\mathbf{I}}(C\,;B)\,.\end{aligned} \tag{3.1}$$

Proof. The chain rules for Shannon and Tsallis information give

$$Q(C) - \mathbf{I}(C:X) - \Delta = \mathbf{I}(C:B\,|\,X) - (1-\bar{\mathrm{I}}_p)\mathbf{I}(B:C) - \mathrm{J}_p\bar{\mathbf{I}}(C\,;B)\,.$$

The chain rule and Lemma 3.4 give

$$\begin{aligned}\mathbf{I}(C:B\,|\,X) &= \mathbf{I}(B:C\,|\,U) - \mathbf{I}(A:C\,|\,U)\\ &\geq (1-\bar{\mathrm{I}}_p)\mathbf{I}(B:C\,|\,U) + \bar{\mathrm{H}}(C\,|\,U)\left[\bar{\mathrm{I}}_p\mathbf{I}(B:C') - \mathbf{I}(A:C')\right].\end{aligned}$$

Substituting this into the preceding identity and applying Lemma 3.6 gives (3.1). □

Lemma 3.8. *If* $\bar{\mathrm{H}}(C\,|\,U) > 0$*, then*

$$Q(C) - \mathbf{I}(C:X) \geq \bar{\mathrm{H}}(C\,|\,U)\,\mathbf{I}(A:B\,|\,C') - \left[\frac{(1-\bar{\mathrm{I}}_p)c(\mu)}{\ln 2} + \mathrm{J}_p\right]\bar{\mathbf{I}}(C\,;U)\,. \tag{3.2}$$

Proof. Using the comparison in $\mathbf{I}(C:X) = \mathbf{I}(C:U) + \mathbf{I}(A:C\,|\,U)$ gives

$$\begin{aligned}Q(C) - \mathbf{I}(C:X) &\geq \bar{\mathrm{I}}_p\mathrm{H}(C) - \mathrm{J}_p\bar{\mathrm{H}}(C) - \mathbf{I}(C:U) - \bar{\mathrm{H}}(C\,|\,U)\,\mathbf{I}(A:C')\\ &= \bar{\mathrm{H}}(C\,|\,U)\,\mathbf{I}(A:B\,|\,C') + \bar{\mathrm{I}}_p\left[\mathrm{H}(C\,|\,U) - \bar{\mathrm{H}}(C\,|\,U)\right]\\ &\quad - (1-\bar{\mathrm{I}}_p)\mathbf{I}(C:U) - \mathrm{J}_p\bar{\mathbf{I}}(C\,;U)\,.\end{aligned}$$

Here we used $\mathrm{I}_p = \mathbf{I}(A:B)$ and $\mathbf{I}(A:B\,|\,C') = \mathbf{I}(A:B) - \mathbf{I}(A:C')$. Now $\mathrm{H}(C\,|\,U) \geq \bar{\mathrm{H}}(C\,|\,U)$ and Lemma 2.2 give (3.2). □

For $0 < p < 1$, we use the shorthand

$$\alpha_p = \frac{\mathrm{J}_p\ln 2}{1-\bar{\mathrm{I}}_p}.$$

We need the following fact about the Markov chain $A - B - C'$ of three unbiased bits. We state it here and prove it in Section 3.3.

Lemma 3.9. *Let C' be an unbiased bit obtained from B using private randomness. If $0 < \bar{\mathrm{I}}_p < 1$ and $\bar{\mathbf{I}}(B\,;C') > 0$, then for every $0 < d \le 1/2$,*

$$\frac{\mathrm{I}_p - \mathbf{I}(A:C')}{2(1+\alpha_p - d/4)} + \frac{\bar{\mathrm{I}}_p \mathbf{I}(B:C') - \mathbf{I}(A:C')}{d\sqrt{\bar{\mathbf{I}}(B\,;C')}} \ge \mathrm{J}_p.$$

Proof of Lemma 3.3. The cases $p \in \{0, 1/2, 1\}$ are immediate, so we may assume $0 < p < 1/2$, complementing Y if necessary. If $\bar{\mathbf{I}}(C\,;B) = 0$, then C and B are independent. Consequently $Q(C) = Q(C\,|\,B)$ and $\mathbf{I}(C:X\,|\,B) \ge \mathbf{I}(C:X)$, proving the claim with $\lambda = 0$.

We may therefore assume $\bar{\mathbf{I}}(C\,;B) > 0$. Then $\bar{\mathrm{H}}(C\,|\,U) > 0$: otherwise C would be determined by U and hence independent of B. By Lemma 3.4, there is an unbiased bit C' obtained from B alone and satisfying (B). Let $d \in (0, 1/2]$ satisfy $d^2 = \bar{\mathbf{I}}(C\,;B)$, and set

$$\lambda = \frac{\bar{\mathbf{I}}(C\,;B)}{2\bar{\mathrm{H}}(C)\;\bar{\mathrm{H}}(C\,|\,U)\,(c(\mu)+\alpha_p)}.$$

All factors are positive. When we multiply (3.2) by λ and add it to (3.1), the terms involving $\bar{\mathbf{I}}(C\,;U)$ cancel, giving

$$\begin{aligned}(1+\lambda)\,(Q(C) - \mathbf{I}(C:X)) \ge \Delta + \bar{\mathrm{H}}(C\,|\,U)\,\big(\bar{\mathrm{I}}_p \mathbf{I}(B:C') - \mathbf{I}(A:C')\big) \\ + \lambda \bar{\mathrm{H}}(C\,|\,U)\;\mathbf{I}(A:B\,|\,C') - \mathrm{J}_p \bar{\mathbf{I}}(C\,;B)\,.\end{aligned} \tag{3.3}$$

By Lemma 3.5,

$$\frac{\bar{\mathrm{H}}(C\,|\,U)}{\bar{\mathbf{I}}(C\,;B)} \ge \frac{1}{d\sqrt{\bar{\mathbf{I}}(B\,;C')}}.$$

Since B is unbiased, $d \le 2\min\{\mu, 1-\mu\}$. Complementing C if necessary, assume $\mu \ge 1/2$, so $\mu \le 1 - d/2$. Using $\ln x \le (x - x^{-1})/2$ for $x \ge 1$ gives

$$\bar{\mathrm{H}}(C)\;c(\mu) = \frac{\mu \ln(1/\mu)}{1-\mu} \le \frac{1+\mu}{2} \le 1 - \frac{d}{4}.$$

Consequently, using $\bar{\mathrm{H}}(C) \le 1$,

$$\frac{\lambda \bar{\mathrm{H}}(C\,|\,U)}{\bar{\mathbf{I}}(C\,;B)} = \frac{1}{2\bar{\mathrm{H}}(C)\,(c(\mu)+\alpha_p)} \ge \frac{1}{2(1+\alpha_p - d/4)}.$$

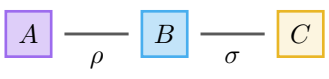
The two information quantities multiplied by these coefficients are nonnegative. For the first, the binary divergence series gives $\mathbf{I}(A:C') \le \bar{\mathbf{I}}(A\,;B)\;\mathbf{I}(B:C')$; the second is a conditional mutual information. Applying Lemma 3.9, we therefore obtain

$$\frac{\bar{\mathrm{H}}(C\,|\,U)}{\bar{\mathbf{I}}(C\,;B)}\,\big(\bar{\mathrm{I}}_p \mathbf{I}(B:C') - \mathbf{I}(A:C')\big) + \frac{\lambda \bar{\mathrm{H}}(C\,|\,U)}{\bar{\mathbf{I}}(C\,;B)}\,\mathbf{I}(A:B\,|\,C') \ge \mathrm{J}_p.$$

Thus the last three terms in (3.3) have nonnegative sum, so $Q(C) - \mathbf{I}(C:X) \ge \Delta/(1+\lambda)$, as required. □

3.3 The three-bit lemma

In this section we study the Markov chain $A - B - C$ of three unbiased bits. The joint law is determined by two parameters: the correlations ρ of A, B and σ of B, C. These satisfy $\rho^2 = \bar{\mathbf{I}}(A\,;B)$ and $\sigma^2 = \bar{\mathbf{I}}(B\,;C)$. Assume $\rho^2 \ne 1$, $\sigma \ne 0$ and fix a $d \in (0, 1/2]$.



To prove Lemma 3.9, we expand the binary divergence $\mathbf{D}_2(x \,\|\, 1/2)$ in a Taylor series about $x = 1/2$. Writing $t = 2x - 1$ for the bias, we define

$$f(t) = \mathbf{D}_2((1+t)/2 \,\|\, 1/2)\,, \qquad \alpha = \frac{(\rho^2 - f(\rho))\ln 2}{1-\rho^2}.$$

Then $\mathbf{I}(A:B) = f(\rho)$, $\mathbf{I}(B:C) = f(\sigma)$, and $\mathbf{I}(A:C) = f(\rho\sigma)$, so it suffices to prove

$$\frac{f(\rho) - f(\rho\sigma)}{2(1+\alpha-d/4)} + \frac{\rho^2 f(\sigma) - f(\rho\sigma)}{d\,|\sigma|} \geq \rho^2 - f(\rho). \tag{3.4}$$

When $\rho = 0$, both sides vanish. Since f is even, it suffices to prove the strict inequality for $\rho, \sigma > 0$. We expand in ρ^2 and prove positivity of every coefficient. For $N \geq 1$, define the divergence tails

$$b_N = \sum_{k>N} \frac{1}{2k(2k-1)} = \int_0^1 \frac{x^{2N}}{1+x}\,\mathrm{d}x,$$
$$s_N(\sigma) = \sum_{k>N} \frac{\sigma^{2k}}{2k(2k-1)} = \int_0^\sigma \frac{(\sigma-x)x^{2N}}{1-x^2}\,\mathrm{d}x.$$

Set $b_0 = 1 - d/4$ as a separate convention, and write

$$B_N = \sum_{i=0}^{N-1} b_i b_{N-i}, \qquad G_N(\sigma) = \sum_{i=0}^{N-1} b_i s_{N-i}(\sigma),$$
$$e_N = \frac{1}{2}\sum_{k=1}^{N} \frac{1}{2k-1}, \qquad C_N(\sigma) = \sum_{k=1}^{N} \frac{1-\sigma^{2k}}{2k(2k-1)}.$$

The first ingredient compares the divergence tails in two ways.

Lemma 3.10. *For $0 < \sigma < 1$ and $N \geq 1$,*

$$G_N(\sigma) \geq B_N \frac{s_N(\sigma)}{b_N}. \tag{3.5}$$

For $N \geq 2$, we also have

$$\frac{s_N(\sigma)}{\sigma b_N} \geq \frac{s_2(\sigma^{N/2})}{\sigma^{N/2} b_2}. \tag{3.6}$$

Proof. First, $s_j(\sigma)/b_j$ is the expectation of the decreasing function $(\sigma - x)_+/(1-x)$ under the probability density proportional to $x^{2j}/(1+x)$ on $(0,1)$. Increasing j tilts this density by an increasing function, so $s_j(\sigma)/b_j$ decreases with j. For $1 \leq j \leq N$, therefore, $s_j(\sigma) \geq b_j s_N(\sigma)/b_N$; summing proves (3.5). Here is the elementary order principle used in this argument. If Z, Z' are independent with the same distribution, then

$$\operatorname{Cov}(f(Z), g(Z)) = \frac{1}{2}\,\mathbb{E}\left[(f(Z) - f(Z'))(g(Z) - g(Z'))\right].$$

It is nonnegative when both functions increase and nonpositive when one increases and the other decreases. Applying this identity to a density ratio gives the stated comparison of expectations. For the second comparison, let us fix $a \geq 0$ and write

$$M_N(a) = \frac{s_N(e^{-a/N})}{e^{-a/N} b_N}.$$

Changing variables in the two tail integrals gives

$$M_N(a) = e^{-2a} \mathop{\mathbb{E}}_{\pi_N}\left[\frac{\sinh(z/N)}{\sinh((z+a)/N)}\right], \qquad \pi_N(\mathrm{d}z) \;\propto \frac{e^{-2z}}{1+e^{z/N}}\,\mathrm{d}z, \quad z > 0. \tag{3.7}$$

The quotient inside the expectation increases with z because coth decreases. It also increases with N, since $x\coth x$ increases for $x > 0$: its derivative has the sign of $\sinh x \cosh x - x > 0$. Moreover, for $K > N$ the density ratio π_K/π_N increases with z, as its logarithmic derivative is

$$\frac{N^{-1}}{1+e^{-z/N}} - \frac{K^{-1}}{1+e^{-z/K}} > 0.$$

The order principle and then the pointwise comparison of quotients show that $M_N(a)$ increases with N. Taking $a = -N\ln\sigma$ gives $e^{-a/N} = \sigma$ and $e^{-a/2} = \sigma^{N/2}$, proving (3.6). All the densities are integrable; at $a = 0$ the quotient in (3.7) equals one. □

We will use the elementary bounds

$$\frac{11}{16} < \ln 2 < \frac{25}{36}.$$

Both follow from $\ln 2 = 2\sum_{j\ge 0}[(2j+1)3^{2j+1}]^{-1}$: keep its first two terms for the lower bound, and replace $2j+1$ by 3 for $j \ge 1$ for the upper bound. The next lemma bounds the two coefficient sums.

Lemma 3.11. *For $N \ge 2$, $0 \le \sigma \le 1$, and $x = 1 - \sigma^{2N}$,*

$$\frac{B_N}{2e_N b_N} \le \frac{11}{10} - \frac{d}{5}, \qquad \frac{C_N(\sigma)}{2e_N b_N} \ge 2x + \frac{2}{3}x^2.$$

Proof. The tail integral, using $1+x \le 2$ and $1+x \ge 2\sqrt{x}$, gives $1/(4N+2) \le b_N \le 1/(4N+1)$. These bounds imply $b_i b_j \le b_{i+j}(b_i + b_j)$ for positive i, j. Also $\sum_{i=1}^{N-1} b_i = e_N + N b_N - \ln 2$. Thus

$$\frac{B_N}{2e_N b_N} \le 1 + \frac{N b_N - (\ln 2 - 1/2) - d/8}{e_N} \le \frac{35}{32} - \frac{3d}{16} \le \frac{11}{10} - \frac{d}{5}.$$

For the middle inequality use $N b_N \le 1/4$, $\ln 2 > 11/16$, $e_N \ge 2/3$, and $d \le 1/2$. The resulting upper numerator $1/16 - d/8$ is nonnegative, so using $e_N \ge 2/3$ preserves the direction of the estimate.

For the second claim, use the first two terms of $1-(1-x)^{k/N}$, the identity $\sum_{k=1}^{N} k/[2(2k-1)] = N/4 + e_N/2$, and $b_N \le 1/(4N+1)$. With $s = 1/N$, these give

$$\frac{C_N(\sigma)}{2e_N b_N} \ge \frac{4+s}{2}\left[x + \left(\frac{1}{2} - \frac{s}{4} - \frac{1}{8e_N}\right)x^2\right].$$

Since $x \ge x^2$, it suffices that

$$\frac{s}{2} + \frac{4+s}{2}\left(\frac{1}{2} - \frac{s}{4} - \frac{1}{8e_N}\right) \ge \frac{2}{3}.$$

At $N = 2$, the left side is $43/64 > 2/3$. For $N \ge 3$, use $e_N \ge 23/30$ and $s \le 1/3$ to bound it below by

$$\frac{31}{46} + \frac{31s}{184} - \frac{s^2}{8} \ge \frac{31}{46} > \frac{2}{3}.$$

□

Lemma 3.12. *For $0 < w < 1$, writing $x = 1 - w^4$,*

$$2x + \frac{2}{3}x^2 + 36\frac{s_2(w)}{w} - 2 > 0.$$

Proof. Keeping just the terms of s_2 indexed by $3 \le k \le 7$ bounds the left side below by the rational polynomial

$$P(w) = \frac{2}{3} - \frac{10}{3}w^4 + \frac{2}{3}w^8 + 36\sum_{k=3}^{7}\frac{w^{2k-1}}{2k(2k-1)}.$$

The ratio

$$\frac{P'(w)}{w^3} = -\frac{40}{3} + \frac{16}{3}w^4 + 18\sum_{k=3}^{7}\frac{w^{2k-5}}{k}$$

is strictly increasing. Also $P'''(w) > 0$ for $w \ge 4/5$: the polynomial part contributes $16w(14w^4 - 5) > 0$, and all tail terms contribute positively. Direct rational arithmetic gives

$$P'(4/5) < 0, \quad P''(4/5) > 18, \quad P(5/6) > 1/128, \quad |P'(5/6)| < 1/4.$$

Taylor's formula therefore gives, for $4/5 \le w < 1$,

$$P(w) \ge P(5/6) + P'(5/6)(w - 5/6) + 9(w - 5/6)^2 > \frac{1}{128} - \frac{1}{576} > 0.$$

On $(0, 4/5]$, monotonicity of $P'(w)/w^3$ makes P decreasing, so $P(w) \ge P(4/5) > 0$ there too. □

Lemma 3.13. *For every $N \geq 1$ and $0 < \sigma \leq 1$,*

$$C_N(\sigma) + \frac{2G_N(\sigma)}{d\sigma} - 2B_N > 0.$$

Proof. First suppose $N \geq 2$ and $\sigma < 1$. By Lemma 3.10, the expression is at least

$$C_N(\sigma) + B_N \left[\frac{2s_N(\sigma)}{d\sigma b_N} - 2\right].$$

If the bracket is nonnegative, we are done. Otherwise apply Lemma 3.11, remembering that an upper bound on B_N gives a lower bound after multiplication by this negative bracket. Put $w = \sigma^{N/2}$ and $x = 1 - w^4$. Since $b_2 = \ln 2 - 7/12 < 1/9$, Lemma 3.10 shows that the expression, divided by $2e_N b_N$, is at least

$$2x + \frac{2}{3}x^2 + \left(\frac{11}{10} - \frac{d}{5}\right)\left[\frac{18s_2(w)}{dw} - 2\right]. \tag{3.8}$$

If $s_2(w)/w \geq 1/198$, this expression is decreasing in $d \in (0, 1/2]$: its derivative is

$$-\frac{99}{5d^2}\frac{s_2(w)}{w} + \frac{2}{5} \leq 0.$$

Its value is therefore at least its value at $d = 1/2$, which is positive by Lemma 3.12.

Suppose instead that $s_2(w)/w < 1/198$. The first two tail terms at $w = 2/3$ already exceed $1/198$, so $w < 2/3$. If $w \leq 3/5$, then $x \geq 6/7$, and dropping the positive terms in the expansion of (3.8) gives the lower bound

$$2x + \frac{2}{3}x^2 - \frac{11}{5} \geq \frac{108}{49} - \frac{11}{5} > 0.$$

For $3/5 < w < 2/3$, we have $x > 4/5$ and $s_2(w)/w > w^5/30 > 1/400$. By AM–GM,

$$\frac{99}{5d}\frac{s_2(w)}{w} + \frac{2d}{5} \geq \sqrt{\frac{792}{25}\frac{s_2(w)}{w}} > \frac{7}{25}.$$

Also $-(18/5)s_2(w)/w > -1/55$. Expanding (3.8) now gives the strict lower bound

$$2 - \frac{11}{5} + \frac{7}{25} - \frac{1}{55} > 0.$$

At $\sigma = 1$, we have $C_N(1) = 0$ and $G_N(1) = B_N$, giving $(2/d - 2)B_N > 0$ for every N. For $N = 1$ and $\sigma < 1$, the coefficient is

$$\frac{1 - \sigma^2}{2} + \left(1 - \frac{d}{4}\right)\left[\frac{2s_1(\sigma)}{d\sigma} - 2b_1\right].$$

If the bracket is nonnegative this is positive. Otherwise, using $1 - d/4 \leq 1$ and $2/d \geq 4$, it is at least $(1 - \sigma^2)/2 + 4s_1(\sigma)/\sigma - 2b_1$. Now $b_1 < 7/36$ and the first two tail terms bound this below by

$$Z(\sigma) = \frac{1}{9} - \frac{\sigma^2}{2} + \frac{\sigma^3}{3} + \frac{2\sigma^5}{15} > 0.$$

Indeed, $Z'(\sigma)/\sigma = -1 + \sigma + 2\sigma^3/3$ increases and is negative at $2/3$, whereas $Z'' > 1$ on $[2/3, 1]$. At $a = 3/4$, $Z(a) = 49/23040 > 1/512$ and $Z'(a) = 3/128 < 1/32$. Taylor's formula gives $Z(\sigma) > 1/512 - 1/2048 > 0$ on $[2/3, 1]$, and monotonicity gives positivity on the remaining interval. □

Lemma 3.9 (restated)**.** *Let $A - B - C$ be a Markov chain of unbiased bits with $\bar{\mathbf{I}}(A\,;B) < 1$ and $\bar{\mathbf{I}}(B\,;C) > 0$. For every $0 < d \leq 1/2$,*

$$\frac{\mathbf{I}(A : B \,|\, C)}{2\left(1 - \frac{d}{4} + \frac{\left(\bar{\mathbf{I}}(A;B) - \mathbf{I}(A:B)\right)\ln 2}{\mathrm{H}(A \,|\, B)}\right)} + \frac{\bar{\mathbf{I}}(A\,;B)\,\mathbf{I}(B:C) - \mathbf{I}(A:C)}{d\sqrt{\bar{\mathbf{I}}(B\,;C)}} \geq \bar{\mathbf{I}}(A\,;B) - \mathbf{I}(A:B)\,.$$

Proof. The divergence-tail definitions give the absolutely convergent expansions

$$\begin{aligned}\alpha &= \sum_{N\geq 1} b_N \rho^{2N},\\ \frac{\ln 2}{1-\rho^2}\left[\rho^2 f(\sigma) - f(\rho\sigma)\right] &= \sum_{N\geq 1} s_N(\sigma)\rho^{2N},\\ \frac{\ln 2}{1-\rho^2}\left[f(\rho) - f(\rho\sigma)\right] &= \sum_{N\geq 1} C_N(\sigma)\rho^{2N}.\end{aligned}$$

Multiplying (3.4), after moving its right side to the left, by $2(1-d/4+\alpha)\ln 2/(1-\rho^2)$ gives

$$\sum_{N\geq 1}\left[C_N(\sigma) + \frac{2G_N(\sigma)}{d\sigma} - 2B_N\right]\rho^{2N} > 0$$

by Lemma 3.13. □

4 Statement on AI use

The proof has been obtained with interactive collaboration with GPT-6 Astra and DeepSeek-V4.1-Flash.

A Remaining proofs

We use the setup of Section 1.3.

Lemma A.1. *For the threshold bit T_μ defined in Section 1.3, we have*

$$Q(T_\mu) \leq \mathbf{I}(T_\mu : X_\circ)\,.$$

For $0 < p < 1/2$, equality holds exactly when $\mu \in \{0, 1/2, 1\}$. At $p = 0, 1/2$, equality holds for every μ.

Proof. The cases $p \in \{0, 1/2\}$ or $\mu \in \{0, 1/2, 1\}$ follow directly. Let us fix $0 < \mu < 1$, $\mu \neq 1/2$, and write

$$\rho = 1 - 2p, \qquad F(\rho) = \mathbf{I}(T_\mu : X_\circ) - \bar{\mathrm{H}}(T_\mu)\,\mathrm{I}_p.$$

We have $F(0) = F'(0) = 0$, $F(1) = \mathrm{H}(T_\mu) - \bar{\mathrm{H}}(T_\mu)$, and differentiation gives

$$F''(\rho) = \frac{\bar{\mathrm{H}}(T_\mu)\,(1-\bar{\mathrm{H}}(T_\mu))(1-\rho)}{\ln 2\,(1+\rho)\big(4\rho + \bar{\mathrm{H}}(T_\mu)\,(1-\rho)^2\big)}.$$

This is strictly decreasing: the numerator decreases and both factors in the denominator increase. Hence, for $0 < \rho < 1$,

$$\frac{F(\rho)}{\rho^2} = \int_0^1 (1-t)F''(\rho t)\,\mathrm{d}t > F(1).$$

Since $\bar{\mathrm{I}}_p = \rho^2$, this is precisely $\mathbf{I}(T_\mu : X_\circ) > Q(T_\mu)$. □

We prove the properties of the comparison function used in Section 3.1 and the estimate used in Lemma 3.5. All endpoint expressions are understood by continuity.

For the following calculations, let us write

$$f(t) = \mathbf{D}_2((1+t)/2 \,\|\, 1/2) = \sum_{k\geq 1} \frac{t^{2k}}{2k(2k-1)\ln 2}, \qquad -1 \leq t \leq 1.$$

Thus f is even, convex, and strictly increasing on $[0, 1]$. Its nonnegative coefficients sum to one, so $f(t) \leq t^2$.

Let A, B be unbiased bits, with absolute correlation ρ. Let C be a nonconstant bit such that $A - B - C$ is a Markov chain, and write

$$V = \bar{\mathrm{H}}(C)\,, \qquad d = \mathbb{E}[(2B-1)(2C-1)].$$

We define F_ρ on $[0,1]$ by

$$F_\rho(f(t)) = f(\rho t), \qquad 0 \le t \le 1.$$

Lemma A.2. *The function F_ρ is increasing and concave, and*

$$\mathbf{I}(B:C) \le V, \qquad \mathbf{I}(A:C) \le V F_\rho\left(\frac{\mathbf{I}(B:C)}{V}\right).$$

Moreover,

$$|d| \le V, \qquad \mathbf{I}(B:C) \ge V f(d/V).$$

Proof. Let us write $x = \mathbb{E}[2C-1]$, so $|x| + |d| \le 1$ and $V = 1 - x^2$. Bayes' rule gives

$$\begin{aligned}\mathbf{I}(B:C) &= \frac{1+x}{2} f\left(\frac{d}{1+x}\right) + \frac{1-x}{2} f\left(\frac{d}{1-x}\right) =: \ell(x,d),\\ \mathbf{I}(A:C) &= \ell(x, \rho d).\end{aligned}$$

Relabeling A if necessary makes its correlation with B nonnegative and does not change either information. Since $f(t) \le t^2$,

$$\ell(x,d) \le \frac{d^2}{V} \le V,$$

where $|d| \le 1 - |x| \le V$ proves the second inequality.

The divergence series also gives

$$\frac{\ell(x,d)}{V} = \sum_{k \ge 1} \frac{W_k(x)(d/V)^{2k}}{2k(2k-1)\ln 2}, \qquad W_k(x) = \frac{(1+x)^{2k-1} + (1-x)^{2k-1}}{2}.$$

Here $W_1 = 1$ and $W_k \ge 1$. This proves $\ell(x,d) \ge V f(d/V)$, as required for the lower bound.

Let us choose $\tau \in [0,1]$ so that $f(\tau) = \ell(x,d)/V$. Cauchy–Schwarz gives $W_k^2 \le W_{k-1} W_{k+1}$ for $k \ge 2$. Together with $W_1 = 1$ and $W_k \ge 1$, this implies that $W_k^{1/k}$ is nondecreasing. Indeed, successive differences of $\ln W_k$ are nondecreasing, so

$$\ln W_{k+1} - \ln W_k \ge \frac{\ln W_k}{k-1} \qquad (k \ge 2).$$

Consequently the terms

$$\frac{W_k(x)(d/V)^{2k} - \tau^{2k}}{2k(2k-1)\ln 2}$$

change sign at most once, from negative to positive. Their absolutely convergent sum is zero by the choice of τ. Multiplication by the decreasing weights ρ^{2k} therefore gives a nonpositive sum: subtracting a weight at the sign change makes each summand nonpositive. Thus

$$\ell(x, \rho d) \le V f(\rho\tau) = V F_\rho\big(\ell(x,d)/V\big).$$

The cases with no sign change, and $\rho = 0, 1$, follow directly.

It remains to prove concavity. For $y = f(t)$ with $0 < t < 1$,

$$F_\rho'(y) = \rho^2 \frac{\sum_{j \ge 0} \rho^{2j} t^{2j}/(2j+1)}{\sum_{j \ge 0} t^{2j}/(2j+1)} \ge 0.$$

This ratio decreases with t: differentiating and pairing terms indexed by j, k gives factors $(j-k)(\rho^{2j} - \rho^{2k}) \le 0$. Since y increases with t, the derivative decreases with y. Continuity gives the assertion on $[0,1]$. □

Finally, the lower pointwise comparison averages in the same way. In the setup of Section 3.1, let us write $v = \bar{\mathrm{H}}(C \,|\, U) > 0$ and $d = \mathbb{E}[(2B-1)(2C-1)]$. For each value u of U, let us write V_u and d_u for the corresponding conditional entropy and correlation. Independence of B and U gives $\mathbb{E}_{u\sim U}\, d_u = d$. Since $|d_u| \le V_u$, we have $|d| \le v$. Jensen's inequality for the convex function f, with weights V_u/v, gives

$$\mathbf{I}(B\!:\!C \,|\, U) \ge \mathop{\mathbb{E}}_{u\sim U}\big[V_u f(d_u/V_u)\big] \ge v f(d/v).$$

Values with $V_u = 0$ have $d_u = 0$ and contribute zero.

B Shannon and Tsallis information

Lemma B.1. *For $a \in [0,1]$ and $b \in [1/2, 1)$,*

$$\frac{2(a-b)^2}{\ln 2} \le \mathbf{D}_2(a \,\|\, b) \le \frac{\ln(1/b)}{(1-b)^2 \ln 2}(a-b)^2.$$

The lower bound holds for every $b \in (0,1)$; complementing a, b gives the corresponding upper bound for $b < 1/2$.

Proof. The second derivative of the divergence in a is $1/(a(1-a)\ln 2) \ge 4/\ln 2$, and its value and first derivative vanish at $a = b$. This proves the lower bound. For the upper bound, Taylor's integral formula gives

$$\frac{\mathbf{D}_2(a \,\|\, b)}{(a-b)^2} = \frac{1}{\ln 2}\int_0^1 \frac{1-t}{(b + t(a-b))(1-b-t(a-b))}\,\mathrm{d}t.$$

The quotient extends continuously to $a = b$ and is convex in a, because $1/(x(1-x)) = 1/x + 1/(1-x)$ is convex. It therefore attains its maximum at an endpoint. For $b \ge 1/2$, the value at $a = 1$ is at least the value at $a = 0$: equivalently,

$$-b^2 \ln b + (1-b)^2 \ln(1-b) \ge 0.$$

The expression vanishes at $b = 1/2$ and $b = 1$, and its second derivative is $2\ln((1-b)/b) \le 0$ on this interval. The value at $a = 1$ is the claimed coefficient. □

For bits, we write $\bar{\mathbf{D}}_2(a \,\|\, b) = 4(a-b)^2$ and $\bar{\mathrm{H}}(a) = 4a(1-a)$. The binary divergence has the expansion

$$\mathbf{D}_2\big(\tfrac{1+u}{2} \,\big\|\, \tfrac{1}{2}\big) = \sum_{k\ge 1} \frac{u^{2k}}{2k(2k-1)\ln 2}, \qquad -1 \le u \le 1. \tag{B.1}$$

Indeed, expanding its second derivative $1/((1-u^2)\ln 2)$ as a geometric series and integrating twice proves the identity for $|u| < 1$; continuity gives the endpoints. The coefficients $c_k = 1/(2k(2k-1)\ln 2)$ are nonnegative and satisfy $\sum_k c_k = 1$ (set $u = 1$). Hence,

$$\mathbf{D}_2\big(\tfrac{1+u}{2} \,\big\|\, \tfrac{1}{2}\big) = \sum_{k\ge 1} c_k u^{2k} \le u^2 = \bar{\mathbf{D}}_2\big(\tfrac{1+u}{2} \,\big\|\, \tfrac{1}{2}\big),$$

with equality iff $u \in \{-1, 0, 1\}$. Since $\mathrm{H}_2(a) = 1 - \mathbf{D}_2(a \,\|\, 1/2)$ and $\bar{\mathrm{H}}(a) = 1 - \bar{\mathbf{D}}_2(a \,\|\, 1/2)$, this gives

$$\bar{\mathrm{H}}(a) \le \mathrm{H}_2(a) \quad (0 \le a \le 1), \qquad \mathbf{I}(B\!:\!Z) \le \bar{\mathbf{I}}(B\,;Z) \quad \text{when } B \text{ is a uniform bit.}$$

In particular $\bar{\mathrm{H}}(B) \le \mathrm{H}(B)$ and $\mathbf{I}(Y_1\!:\!X) = \mathrm{I}_p \le \bar{\mathrm{I}}_p = \bar{\mathbf{I}}(Y_1\,;X)$, the orderings used in Figure 1. The second inequality needs the uniform reference: for a nonconstant biased bit B, taking $Z = B$ gives $\mathbf{I}(B\!:\!B) = \mathrm{H}(B) > \bar{\mathrm{H}}(B) = \bar{\mathbf{I}}(B\,;B)$.

Proof of Lemma 2.1. Let us write $\mu_Z = \Pr[C = 1 \mid Z]$ and $\mu = \mathbb{E}\, C$. Expanding the definition gives

$$\bar{\mathbf{I}}(C\,;Z) = 4\,\mathbb{E}\big[(\mu_Z - \mu)^2\big].$$

This proves nonnegativity, while the upper bound by $\bar{\mathrm{H}}(C)$ follows from $\bar{\mathrm{H}}(C \mid Z) \geq 0$. Both statements may be averaged over U. We already have $\bar{\mathrm{H}}(C) \leq \mathrm{H}(C)$. The lower bound of Lemma B.1 with reference mean $1/2$ also gives

$$1 - \mathrm{H}(C) = \mathbf{D}_2(\mu \,\|\, 1/2) \geq \frac{1 - \bar{\mathrm{H}}(C)}{2\ln 2} \geq \frac{1 - \bar{\mathrm{H}}(C)}{2}.$$

Averaging over U completes part 1.

The chain rule follows by telescoping the entropy differences. If $C - Z - W$ is a Markov chain, then $\bar{\mathrm{H}}(C \mid Z, W) = \bar{\mathrm{H}}(C \mid Z)$, so the chain rule also gives the data processing inequality in part 2.

The lower bound of Lemma B.1, averaged over the conditional mean μ_Z, gives

$$\mathbf{I}(C : Z) \geq \frac{2\,\mathbb{E}[(\mu_Z - \mu)^2]}{\ln 2} = \frac{\bar{\mathbf{I}}(C\,;Z)}{2\ln 2}.$$

The upper comparison for unbiased C was proved above using (B.1). Averaging both comparisons over U gives part 3; the upper one requires C to remain unbiased given U. If a conditional law of C is constant, both informations are zero.

For two bits, evaluating the two conditional means gives

$$\bar{\mathrm{H}}(C)\;\bar{\mathbf{I}}(B\,;C) = 16\,\mathrm{Cov}(B, C)^2 = \bar{\mathrm{H}}(B)\;\bar{\mathbf{I}}(C\,;B)\,.$$

If either bit is constant, all three terms vanish. When B is unbiased, the definition also gives

$$\bar{\mathbf{I}}(C\,;B) = \big(\Pr[C = 1 \mid B = 1] - \Pr[C = 1 \mid B = 0]\big)^2 = \big(2\Pr[B = C] - 1\big)^2.$$

This proves part 4.

For part 5, let us write $V_u = \bar{\mathrm{H}}(C \mid U = u)$ and $\sigma_u = \mathrm{Cov}(B, C \mid U = u)$. The identity for two bits, applied conditionally, and Cauchy–Schwarz give

$$\begin{aligned}\bar{\mathbf{I}}(B\,;C \mid U)\;\bar{\mathrm{H}}(C \mid U) &= 16\left(\mathop{\mathbb{E}}_{u\sim U}\frac{\sigma_u^2}{V_u}\right)\left(\mathop{\mathbb{E}}_{u\sim U} V_u\right)\\ &\geq 16\left(\mathop{\mathbb{E}}_{u\sim U}\sigma_u\right)^2.\end{aligned}$$

Terms with $V_u = 0$ have $\sigma_u = 0$ and contribute zero. Since B is independent of U, its conditional mean is constant, and hence $\mathbb{E}_{u\sim U}\,\sigma_u = \mathrm{Cov}(B, C)$. The last display is therefore at least $\bar{\mathrm{H}}(B)\;\bar{\mathbf{I}}(C\,;B)$, as required.

Finally, applying the Shannon comparisons to the unbiased noise bits gives $\bar{\mathrm{I}}_p/(2\ln 2) \leq \mathrm{I}_p \leq \bar{\mathrm{I}}_p$. Since $\ln 2 < 1$, we have $\bar{\mathrm{I}}_p \leq 2\mathrm{I}_p$, proving part 6. □

Proof of Lemma 2.2. Complementing C if necessary lets us assume $\mu \geq 1/2$. Averaging the upper bound of Lemma B.1 over the conditional mean $\Pr[C = 1 \mid U]$ gives the first inequality. For the second,

$$\bar{\mathrm{H}}(C)\;c(\mu) = \frac{\mu}{1-\mu}\ln\left(1 + \frac{1-\mu}{\mu}\right) \leq 1.$$

using $\ln(1 + x) \leq x$. □

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